Foundations of Named entity recognition

[Named entity recognition]

1.0 Content Block

Definition In the expression named entity, the word named restricts the task to those entities for which one or many strings, such as words or phrases, stand (fairly) consistently for some referent. This is closely related to rigid designators, as defined by Saul Kripke, although in practice NER deals with many names and referents that are not philosophically "rigid". For instance, the automotive company created by Henry Ford in 1903 can be referred to as Ford or Ford Motor Company, although "Ford" can refer to many other entities as well (see Ford). Rigid designators include proper names as well as terms for certain biological species and substances, but exclude pronouns (such as "it"; see coreference resolution), descriptions that pick out a referent by its properties (see also De dicto and de re), and names for kinds of things as opposed to individuals (for example "Bank"). Full named-entity recognition is often broken down, conceptually and possibly also in implementations, as two distinct problems: detection of names, and classification of the names by the type of entity they refer to (e.g. person, organization, or location). The first phase is typically simplified to a segmentation problem: names are defined to be contiguous spans of tokens, with no nesting, so that "Bank of America" is a single name, disregarding the fact that inside this name, the substring "America" is itself a name. This segmentation problem is formally similar to chunking. The second phase requires choosing an ontology by which to organize categories of things. Temporal expressions and some numerical expressions (e.g., money, percentages, etc.) may also be considered as named entities in the context of the NER task. While some instances of these types are good examples of rigid designators (e.g., the year 2001) there are also many invalid ones (e.g., I take my vacations in “June”). In the first case, the year 2001 refers to the 2001st year of the Gregorian calendar. In the second case, the month June may refer to the month of an undefined year (past June, next June, every June, etc.). It is arguable that the definition of named entity is loosened in such cases for practical reasons. The definition of the term named entity is therefore not strict and often has to be explained in the context in which it is used. Certain hierarchies of named entity types have been proposed in the literature. BBN categories, proposed in 2002, are used for question answering and consists of 29 types and 64 subtypes. Sekine's extended hierarchy, proposed in 2002, is made of 200 subtypes. More recently, in 2011 Ritter used a hierarchy based on common Freebase entity types in ground-breaking experiments on NER over social media text.

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Named-entity recognition (NER) (also known as (named) entity identification, entity chunking, and entity extraction) is a subtask of information extraction that seeks to locate and classify named entities mentioned in unstructured text into pre-defined categories such as person names (PER), organizations (ORG), locations (LOC), geopolitical entities (GPE), vehicles (VEH), medical codes, time expressions, quantities, monetary values, percentages, etc. Most research on NER/NEE systems has been structured as taking an unannotated block of text, such as transducing:

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Current challenges Despite high F1 numbers reported on the MUC-7 dataset, the problem of named-entity recognition is far from being solved. The main efforts are directed to reducing the annotations labor by employing semi-supervised learning, robust performance across domains and scaling up to fine-grained entity types. In recent years, many projects have turned to crowdsourcing, which is a promising solution to obtain high-quality aggregate human judgments for supervised and semi-supervised machine learning approaches to NER. Another challenging task is devising models to deal with linguistically complex contexts such as Twitter and search queries. There are some researchers who did some comparisons about the NER performances from different statistical models such as HMM (hidden Markov model), ME (maximum entropy), and CRF (conditional random fields), and feature sets. And some researchers recently proposed graph-based semi-supervised learning model for language specific NER tasks. A recently emerging task of identifying "important expressions" in text and cross-linking them to Wikipedia can be seen as an instance of extremely fine-grained named-entity recognition, where the types are the actual Wikipedia pages describing the (potentially ambiguous) concepts. Below is an example output of a Wikification system:

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GATE supports NER across many languages and domains out of the box, usable via a graphical interface and a Java API. OpenNLP includes rule-based and statistical named-entity recognition. spaCy features fast statistical NER as well as an open-source named-entity visualizer. Hand-crafted grammar-based systems typically obtain better precision, but at the cost of lower recall and months of work by experienced computational linguists. Statistical NER systems typically require a large amount of manually annotated training data. Semisupervised approaches have been suggested to avoid part of the annotation effort. In the statistical learning era, NER was usually performed by learning a simple linear regression model on engineered features, then decoded by a bidirectional Viterbi algorithm. Some commonly used features include:

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Formal evaluation To evaluate the quality of an NER system's output, several measures have been defined. The usual measures are called precision, recall, and F1 score. However, several issues remain in just how to calculate those values. These statistical measures work reasonably well for the obvious cases of finding or missing a real entity exactly; and for finding a non-entity. However, NER can fail in many other ways, many of which are arguably "partially correct", and should not be counted as complete success or failures. For example, identifying a real entity, but:

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[Jim]Person bought 300 shares of [Acme Corp.]Organization in [2006]Time. In this example, a person name consisting of one token, a two-token company name and a temporal expression have been detected and classified.

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Difficulties NER can have reference resolution ambiguities where the same name can refer to different entities of the same type. For example, "JFK" can refer to the former president or his son. The same name can refer to completely different types. "JFK" might refer to the airport in New York. "IRA" can refer to Individual Retirement Account or International Reading Association. This can be caused by metonymy. For example, "The White House" can refer to an organization instead of a location.